

Abhay Sahu

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SUMMARY

Computer Science undergraduate (CSE'27) specializing in **scalable backend systems, microservices architecture, and AI-driven applications**. Experienced in building real-time event-driven systems, deploying ML models into production, and integrating Explainable AI (XAI). Strong foundation in **Java, Spring Boot, FastAPI, and cloud-native infrastructure**. Passionate about solving real-world problems through high-performance and intelligent systems.

PROJECTS

SecureWealth AI – Event-Driven Fraud Detection Engine

Python, FastAPI, Kafka, Docker

- Architected a real-time fraud detection microservice using **XGBoost**, achieving **sub-30ms transaction scoring latency**.
- Integrated **Explainable AI (SHAP)** to generate human-readable insights for flagged transactions (e.g., device mismatch, unusual timing).
- Designed an **event-driven architecture** using Apache Kafka for asynchronous transaction streaming.
- Built scalable infrastructure with **PostgreSQL (persistence)** and **Redis (caching)** in a Dockerized environment.
- Developed comprehensive **Pytest suite** ensuring model accuracy (AUC: 1.0), API reliability, and system performance.

SafeSphere – AI Disaster Preparedness Platform

Django, JavaScript, Gemini API

- **National Hackathon Winner** – 1st place at Anveshna Hackathon for AI-based disaster response solution.
- Integrated **Gemini API** for context-aware safety recommendations using prompt engineering and rule-based orchestration.
- Built features like **safety maps, SOS alerts, safe-route navigation, and interactive drills**.
- Developed 3D learning simulations using **Three.js** and integrated real-time APIs like Mapbox.

BharatStudio – AI Content Generation Platform

- Developed a generative AI-based platform for automated content and media creation in hackathon environments.
- Focused on **AI orchestration workflows** for rapid content generation and deployment.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript

Backend: Spring Boot, FastAPI, Django, Hibernate, JDBC

Databases: PostgreSQL, MySQL, MongoDB, Redis

Cloud & DevOps: Docker, Apache Kafka, OCI, CI/CD

AI/ML: XGBoost, SHAP (Explainable AI), Gemini API, Prompt Engineering, RAG

Frontend: React.js, HTML, CSS, Three.js

Core CS: DSA, System Design, REST APIs

EDUCATION

Sagar Institute of Research & Technology, Bhopal

2023 – 2027

Bachelor of Technology in Computer Science

CGPA: 7.7 / 10

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Developer Professional

Oracle Cloud Infrastructure 2025 Generative AI Professional

MySQL HeatWave Implementation Certified Associate

NPTEL – Java Programming (Elite), Machine Learning

ACHIEVEMENTS

1st Prize – Anveshna National Hackathon (AI Social Impact Solution)

1st Rank – College Data Structures & Algorithms Competition

Solved 150+ LeetCode and 100+ GeeksforGeeks problems

Active technical leader driving innovation and project execution